Venkata Lakshmi Parimala Pasupuleti

Senior Innovation Engineer · PropTech · Real Estate SaaS · Document Intelligence · LLM Reasoning · Agent Harnesses · RAG Pipelines · Eval Tooling · Python · Claude · OpenAI · LangGraph

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PROFESSIONAL SUMMARY

8 years in real estate / PropTech (Entrata) building production backend systems for property management, compliance workflows, and financial settlement at 100K+ MAU scale. Built LeaseI — document extraction and LLM reasoning over unstructured lease PDFs (legal clauses, covenants, compliance rules) with an 8-agent LangGraph harness, HITL escalation, and eval infrastructure — the same document-reasoning problem Built solves for inspection reports and construction documents. Built AgentProbe, an open-source LLM eval toolkit matching Built's internal eval tooling use case: 5-metric engine, CI-integrated regression harness, model routing across Claude/OpenAI/Groq/Ollama. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Building ML models and data pipelines; surfacing business insights into application-facing APIs and reporting layers.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic mission-planning system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) with RAG over regulatory corpus, validation guardrails, and grounded compliance reasoning — shipped to real client, iterated weekly on feedback.

George Mason University
Aug 2024 – Aug 2025

Research Associate · College of Engineering and Computing

- Built prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest anomaly detection across 3 research pipelines; fine-tuned HuggingFace classification model with PyTorch, deployed inference API on Quart.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned compliance and financial settlement systems for 7 years** — annual HUD income-limit pipeline (few thousand federal rows → ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units); financial data correctness is non-negotiable in real estate; zero compliance violations across the ownership period.
- **Built RabbitMQ-based async processing system** for account state updates across thousands of tenant records — retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents. Async orchestration patterns (Celery, Airflow also in stack) applied to high-stakes financial workflows.
- **Owned multi-tenant PostgreSQL schemas** for eligibility, financial, and document workflows — optimized 200+ SQL queries (30% gain); designed for data isolation and consistency across 100K+ MAU in a multi-tenant SaaS environment.
- **Led AWS S3 document migration** — inventoried document types, mapped legacy records to target APIs, validated migrated content and metadata, confirmed all workflows remained accurate post-cutover; 100% PHI compliance, zero security incidents.
- Shipped 20+ full-stack modules across 120+ Agile sprints; led microservices redesign (Docker) — 50% scalability improvement, 40% deployment time reduction; eliminated 80% of manual operational workload through automation pipelines.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built property management modules (Python, Java, PostgreSQL); led AWS S3 cloud migration; redesigned document storage architecture migrating hundreds of thousands of records with full data consistency validation.

PROJECTS

Leasel: Document Intelligence + LLM Reasoning Platform
DOCUMENT EXTRACTION · LANGGRAPH · PROPTECH

Python · LangGraph (8-agent) · FastAPI · TF-IDF + Semantic RAG · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Document extraction over 1,240 unstructured lease PDFs — OCR-fallback ingestion, section-aware chunking, hybrid TF-IDF + semantic search, reranking, citation grounding to exact clause. Same problem as inspection reports and construction covenants: non-standardized documents, clause extraction, conflicting-clause detection, compliance reasoning.
- 8-agent LangGraph harness (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional HITL routing — code-enforced rule: LLM extracts and reasons; human or deterministic logic makes every final regulated call.
- Eval results across 10 compliance rules: 93.6% precision, 87.9% F1; pass@1 81.7%, pass@3 88.3% via multi-trial sampling with LLM-as-judge scoring and CI regression gate.

AgentProbe: LLM Eval Tooling
EVAL FRAMEWORK · CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI eval backend: 5-metric engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, model routing across Claude/OpenAI/Groq/Ollama, offline + online evaluation loops. Used to evaluate LeaseI across 1,240+ compliance test cases.

AI on FHIR: NLP-Driven Medical Records Platform
LLM + BERT · HIPAA

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings over structured records; HIPAA-compliant RBAC; reduced manual retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, Scala

AI & Agents: LangChain, LangGraph, Agent Harnesses, Multi-Agent Orchestration, RAG Pipelines, Document Extraction, Document Intelligence, OCR, Section-Aware Chunking, Hybrid Search, Semantic Search, Citation Grounding, LLM Evaluation, LLM-as-Judge, Golden Datasets, Regression Harnesses, Hallucination Detection, Model Routing, Prompt Engineering, Guardrails, HITL, Structured Outputs, Cost Controls, Embeddings, ChromaDB, Qdrant, OpenAI API, Anthropic API (Claude), Groq, Ollama

Backend & Distributed: FastAPI, REST APIs, Spring Boot, Django, Microservices, Distributed Systems, RabbitMQ, Celery, Temporal, Apache Airflow, Apache Kafka, Redis, Idempotency, API Design, Multi-Tenant Architecture

Cloud & Infra: AWS (S3, EC2, RDS, Glue, Lambda, Bedrock, SageMaker), Azure (Databricks, Data Factory, ADLS, AML), Docker, Kubernetes, Terraform, Pulumi, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry

Data: PostgreSQL, Snowflake, Elasticsearch, MySQL, ETL, Lakehouse Architecture, Data Governance, Query Optimization, NumPy, Pandas, PyTorch, HuggingFace, scikit-learn, XGBoost